

Introduction to Volatility Models

ARCH and GARCH: why return sizes cluster

Clustering



ARCH



GARCH



EWMA · SV

Five steps build the big picture of volatility modeling

From basic concepts to GARCH, then a comparison with its neighbors

01	————	The volatility problem	Why model volatility at all Returns · volatility · clustering
02	————	The ARCH model	Yesterday's shock sets today's variance ARCH(1) · properties · ARCH(q) limits
03	————	The GARCH model	Floor, news and memory GARCH(1,1) · long run · forecast · MLE
04	————	GARCH vs. EWMA	Mean reversion or none Weights · RiskMetrics · COVID
05	———	GARCH vs. SV · extensions	Is volatility fixed or random? Timeline · estimation · GJR · EGARCH

Clustering → ARCH → GARCH → compare with EWMA and SV → extensions

01

The Volatility Problem

We cannot predict the direction of returns, but we can partly predict their size

Volatility measures how far returns typically stray from their mean

Basic terms — log return, standard deviation, annualization

Return — how much today's price moved versus yesterday's

$$r_t = \ln P_t - \ln P_{t-1} \approx \frac{P_t - P_{t-1}}{P_{t-1}}$$

Volatility — standard deviation of returns, scaled to one year (≈ 252 trading days)

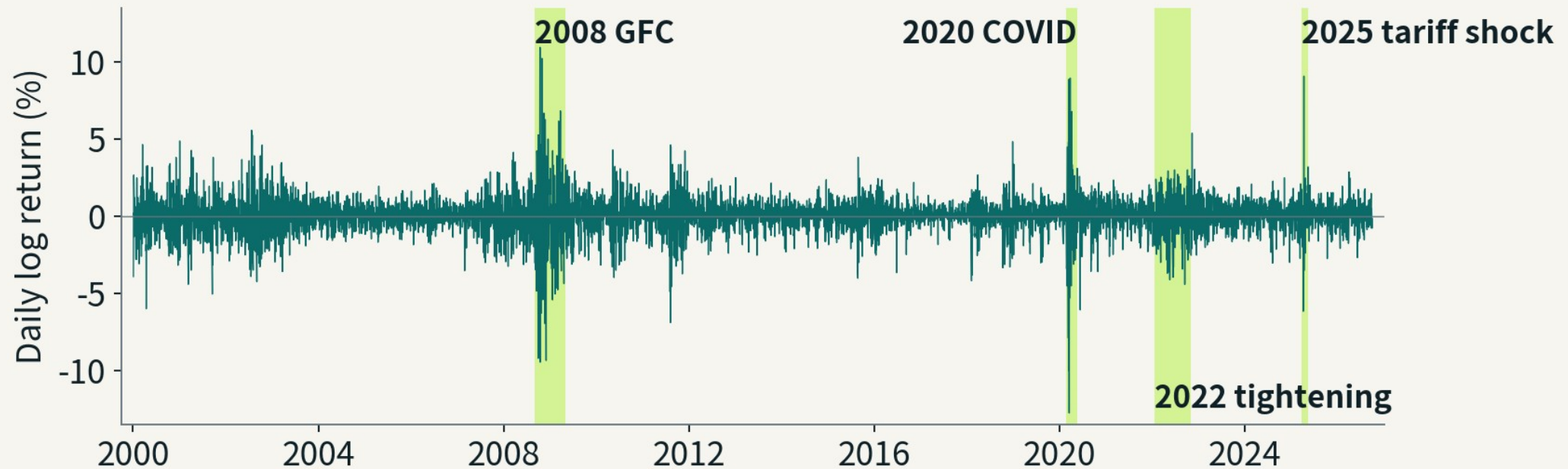
$$\sigma = \sqrt{\frac{1}{n-1} \sum_{t=1}^n (r_t - \bar{r})^2}, \quad \sigma_{\text{annual}} = \sigma_{\text{daily}} \times \sqrt{252}$$

20% volatility means a typical year lands within about $\pm 20\%$ of the average return

※ S&P 500 daily standard deviation $1.21\% \times \sqrt{252} \approx 19.2\%$ (2000–2026)

Large moves arrive in clusters of large moves

S&P 500 daily log returns, Jan 2000 – Sep 2026; shaded bands mark major stress episodes



※ This is volatility clustering — calm periods and turbulent periods take turns

Returns are hard to predict, but their magnitude is predictable

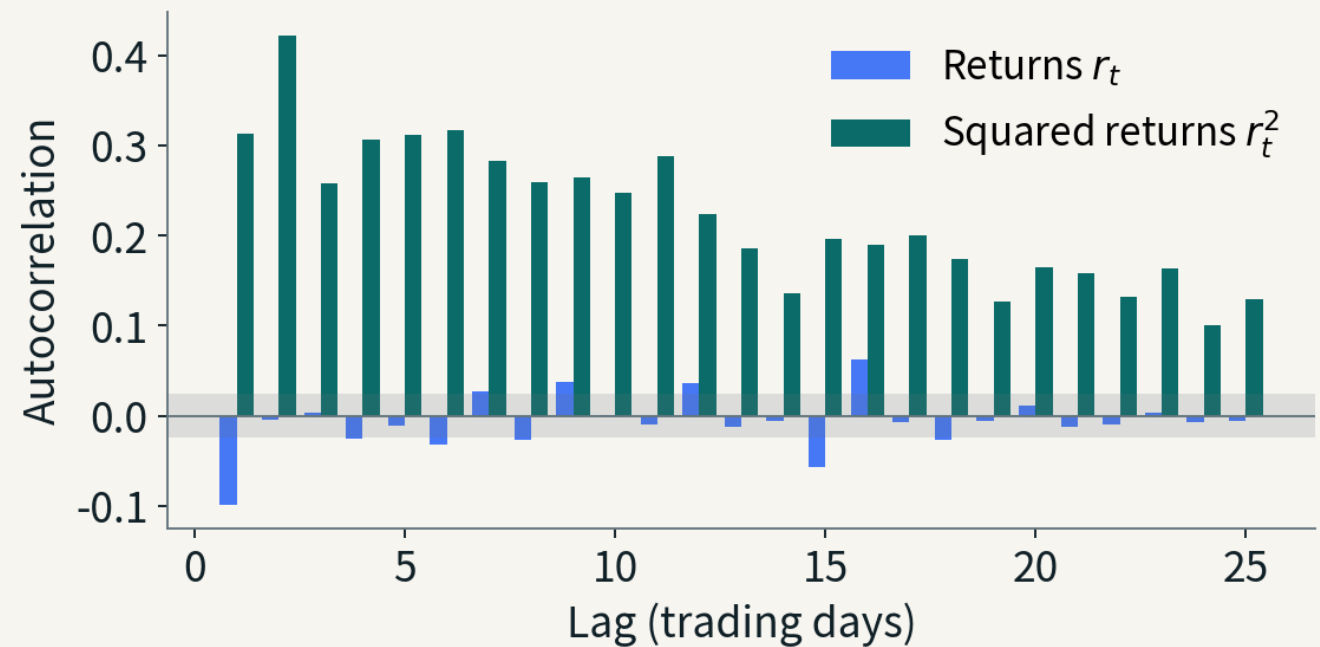
Autocorrelation of S&P 500 daily returns and squared returns; grey band = 95% bounds

Key evidence

Size has memory

- Return autocorr. ≈ 0
- $r^2 > 0$ for 25 days

Magnitude can be modeled



Real returns have far fatter tails than the normal distribution

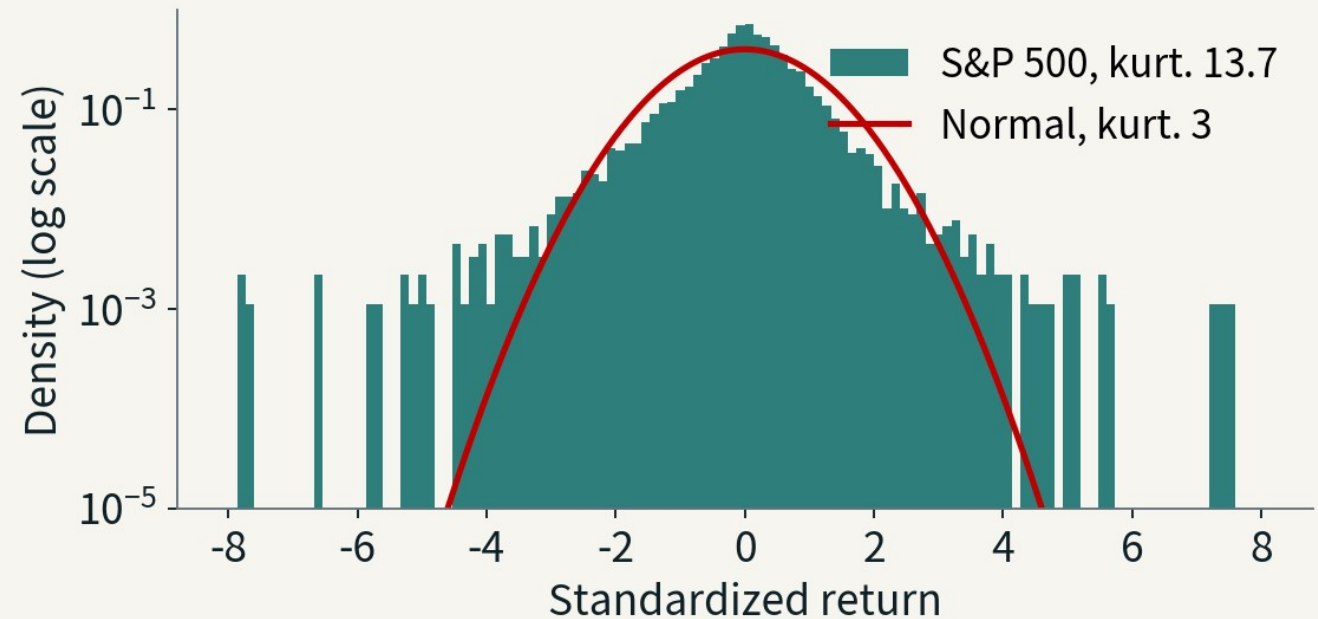
Distribution of standardized S&P 500 daily returns, log vertical scale

Fat tails · normal = 3

Kurtosis 13.7

- 24 moves beyond $\pm 5\sigma$
- Normal predicts 0.004

**Changing volatility
fattens tails**



Volatility models target the conditional variance

$\mathcal{F}(t-1)$ denotes everything observed up to yesterday's close

Unconditional variance — average wobble over the whole sample (one constant)

$$\text{Var}(r_t) = E \left[(r_t - E[r_t])^2 \right] = \sigma^2$$

Conditional variance — expected wobble today, given what we knew yesterday (changes daily)

$$\text{Var}(r_t \mid \mathcal{F}_{t-1}) = E \left[(r_t - E[r_t \mid \mathcal{F}_{t-1}])^2 \mid \mathcal{F}_{t-1} \right] = \sigma_t^2$$

Textbook variance is unconditional; the σ_t^2 of volatility models is conditional

ARCH and GARCH are rules for how σ_t^2 changes from day to day

In time series, heteroskedasticity is a target, not a defect

Heteroskedasticity = the error variance differs across observations

Cross-section · $\text{Var}(\varepsilon \mid x)$

A nuisance

- Biases standard errors
- Fixed with White SEs

Something to correct



Financial time series · $\text{Var}(\varepsilon \mid \text{past})$

Information

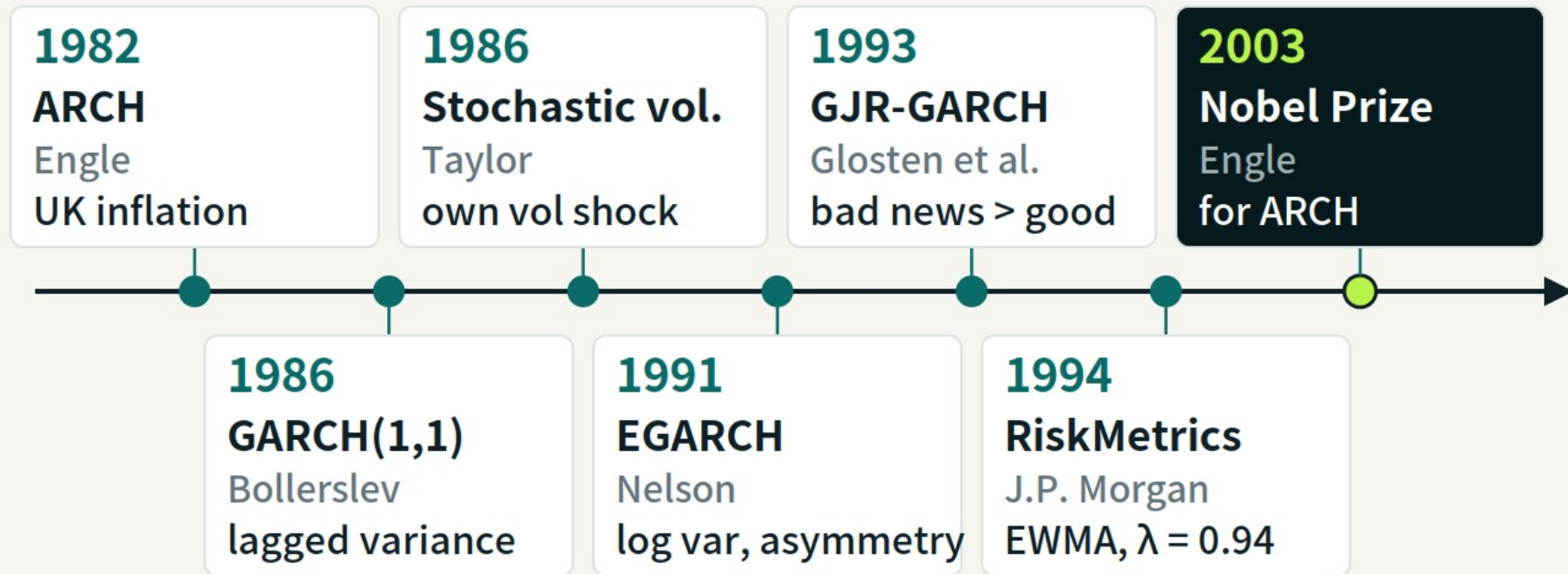
- Risk changes over time
- Forecast tomorrow's risk

Something to model

Before the 1980s econometrics modeled the mean; afterwards it began to model the variance itself

Volatility modeling began with Engle (1982) and became industry standard

Lineage of key models — the academic ARCH family, industry EWMA, and the separate SV branch



※ Engle shared the 2003 Nobel Prize in Economics with Granger, cited for the ARCH model

Volatility forecasts feed risk management, derivatives and allocation

One volatility number enters many decisions at a financial institution at once

Risk mgmt

VaR · ES

Worst-case loss

Derivatives

Options

Vol = price

Allocation

Vol targeting

Sizes positions

Common input

σ_t forecast

Feeds them all

Example: a fund targeting 10% volatility halves its equity weight when forecast volatility hits 20%

A bad forecast either understates risk or forgoes opportunity

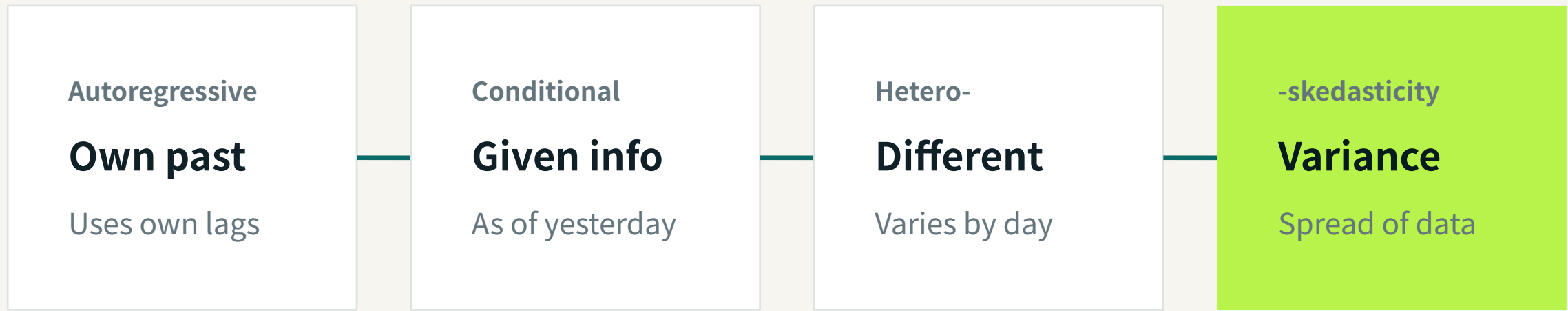
02

The ARCH Model

Yesterday's shock sets today's variance — Engle (1982)

ARCH means a variance that changes by leaning on its own past

Autoregressive Conditional Heteroskedasticity — unpacked one word at a time



Together: a day-to-day variance, set by yesterday's information and explained by its own past

ARCH applies an AR model to the variance instead of the mean

A squared return is the simplest one-day estimate of variance

Daily mean returns are near zero — with one observation, the variance formula collapses to r^2

Shrink the sample-variance formula to a single day

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (r_i - \bar{r})^2 \xrightarrow{n=1, \bar{r} \approx 0} \hat{\sigma}_t^2 = r_t^2$$

Pro

Observed daily

Ready at the close

Con

Very noisy

One day is erratic

ARCH's idea

Blend many days

Weighted past r^2

ARCH = build tomorrow's variance from squared returns up to today

ARCH(1) sets today's variance from yesterday's squared return

First-order ARCH — z_t is white noise with mean 0 and variance 1

ARCH(1)

$$r_t = \sigma_t z_t, \quad z_t \sim N(0, 1), \quad \sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2$$

$$\alpha_0 > 0$$

Floor

Variance on a calm day

$$\alpha_1 \geq 0$$

Reaction

Shock pass-through

z_t

Direction

Fresh random draw daily

A big move yesterday raises today's variance, so another big move becomes more likely

With the same average variance, ARCH creates calm spells and bursts

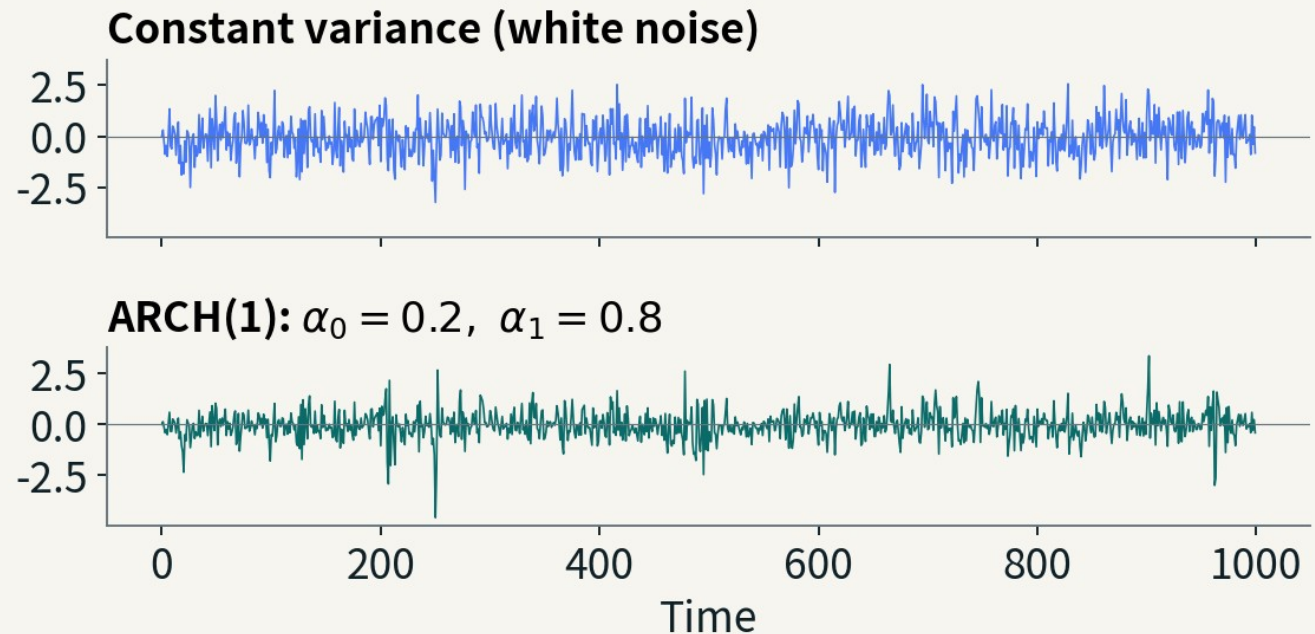
Simulation of 1,000 days — both series have a long-run variance of 1

Simulation

Shocks breed shocks

- Top: constant variance
- Bottom: ARCH(1)

Clustering and fat tails appear together



ARCH(1) has a constant long-run variance and fatter-than-normal tails

Conditional variance moves daily, yet the unconditional variance converges to one constant

Unconditional variance and kurtosis (stationarity: $0 \leq \alpha_1 < 1$)

$$E[r_t^2] = \frac{\alpha_0}{1 - \alpha_1}, \quad \text{Kurt}(r_t) = 3 \frac{1 - \alpha_1^2}{1 - 3\alpha_1^2} > 3$$

Stationarity

$$\alpha_1 < 1$$

Shocks eventually fade

Finite 4th moment

$$3\alpha_1^2 < 1$$

Kurtosis stays finite

Result

Kurtosis > 3

Tails get fatter

Even with normal z_t , r_t has fat tails — part of the 13.7 kurtosis we saw earlier

Real data demand long lags, which makes ARCH(q) heavy

Volatility clusters last for weeks, not days — recall the autocorrelation chart

ARCH(q)

$$\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + \alpha_2 r_{t-2}^2 + \cdots + \alpha_q r_{t-q}^2$$

Problem with ARCH(q)

Many unstable parameters

q of 10–20, all must be ≥ 0

Bollerslev's fix

Add one lagged variance

3 parameters, long memory → GARCH

ARCH(1) on the S&P 500 gives $\alpha_1 = 0.39$ — one day of memory cannot capture the clustering

A volatility model is a pair: a mean equation and a variance equation

The mean equation handles direction; the variance equation handles size

Mean equation — e.g. AR(1); a constant alone is most common

$$r_t = \mu + \phi r_{t-1} + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t$$

Variance equation — ARCH(1) or GARCH(1,1)

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 \quad \text{or} \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

“Estimating a GARCH model” means estimating both equations jointly

※ ARCH-M puts the variance into the mean equation ($r_t = \mu + \lambda \sigma_t^2 + \varepsilon_t$) — higher risk, higher expected return

03

The GARCH Model

Adding lagged variance compresses ARCH's long lags into three parameters

GARCH(1,1) builds today's variance from floor, news and memory

Bollerslev (1986) — if ARCH is like an AR model, GARCH is like an ARMA model

GARCH(1,1)

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

ω — floor

Constant

Pull toward the long run

α — news

Yesterday's shock

Response to news

β — memory

Yesterday's variance

Sums up the past

S&P 500 estimates: $\alpha = 0.120$, $\beta = 0.860$ — memory (β) is about $7\times$ news (α)

GARCH(1,1) is an ARCH(∞) with only three parameters

Substitute the equation into σ_{t-1}^2 repeatedly and you get a weighted sum of all past shocks

After repeated substitution

$$\sigma_t^2 = \frac{\omega}{1 - \beta} + \alpha \sum_{k=1}^{\infty} \beta^{k-1} \varepsilon_{t-k}^2$$

Weight on a k-day-old shock

$$\alpha \cdot \beta^{k-1}$$

Decays geometrically

S&P 500, 20 days back

0.0069

$\approx$ 6% of yesterday's

Takeaway

Long memory

Only 3 parameters

In practice a single GARCH(1,1) describes most equity indices well

$\alpha + \beta$ sets shock persistence; long-run variance sets where volatility returns

If $\alpha + \beta < 1$, volatility eventually reverts to its long-run level (mean reversion)

Long-run variance and half-life — days until half of a shock has faded

$$V_L = \frac{\omega}{1 - \alpha - \beta}, \quad \text{half-life} = \frac{\ln 0.5}{\ln(\alpha + \beta)}$$

$\alpha + \beta$

0.981

Long-run vol (ann.)

18.1%

Half-life

35 days

Latest vol (Sep 2026)

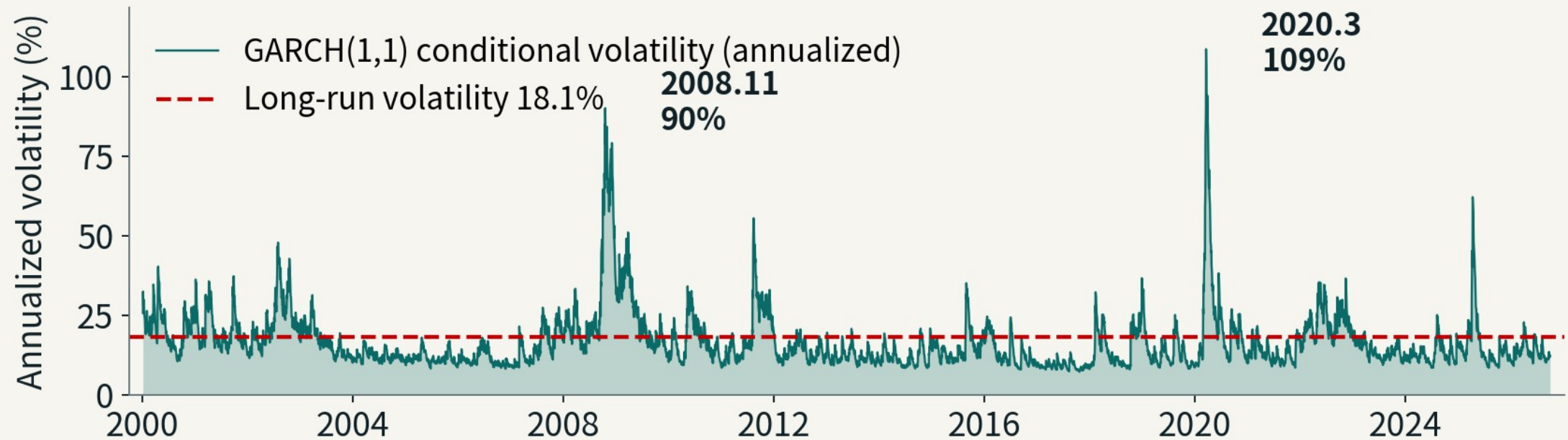
12.2%

Volatility now sits below its long-run level, so GARCH forecasts a gradual rise

※ GARCH(1,1), Gaussian maximum likelihood, S&P 500 daily log returns (Jan 2000 – Sep 2026)

S&P 500 GARCH volatility spikes in crises, then reverts

GARCH(1,1) conditional volatility annualized by $\sqrt{252}$, Jan 2000 – Sep 2026



※ Peaks: Nov 2008 $\approx$ 90%, Mar 2020 $\approx$ 109% — normal times run at 10–15%

GARCH forecasts approach the long-run variance as the horizon grows

Expected variance k days ahead — the gap to the long run shrinks by $(\alpha + \beta)^{k-1}$

k-step-ahead variance forecast

$$E_t \left[\sigma_{t+k}^2 \right] = V_L + (\alpha + \beta)^{k-1} \left(\sigma_{t+1}^2 - V_L \right)$$

k = 1

Tomorrow

Direct from the equation

k grows

Gap shrinks

$(\alpha + \beta)^{k-1} \rightarrow 0$

k → ∞

Long-run variance

Converges to V_L

A 30-day average volatility averages the 1- to 30-day forecasts — matched to option maturities

High and low volatility both converge to the long-run level

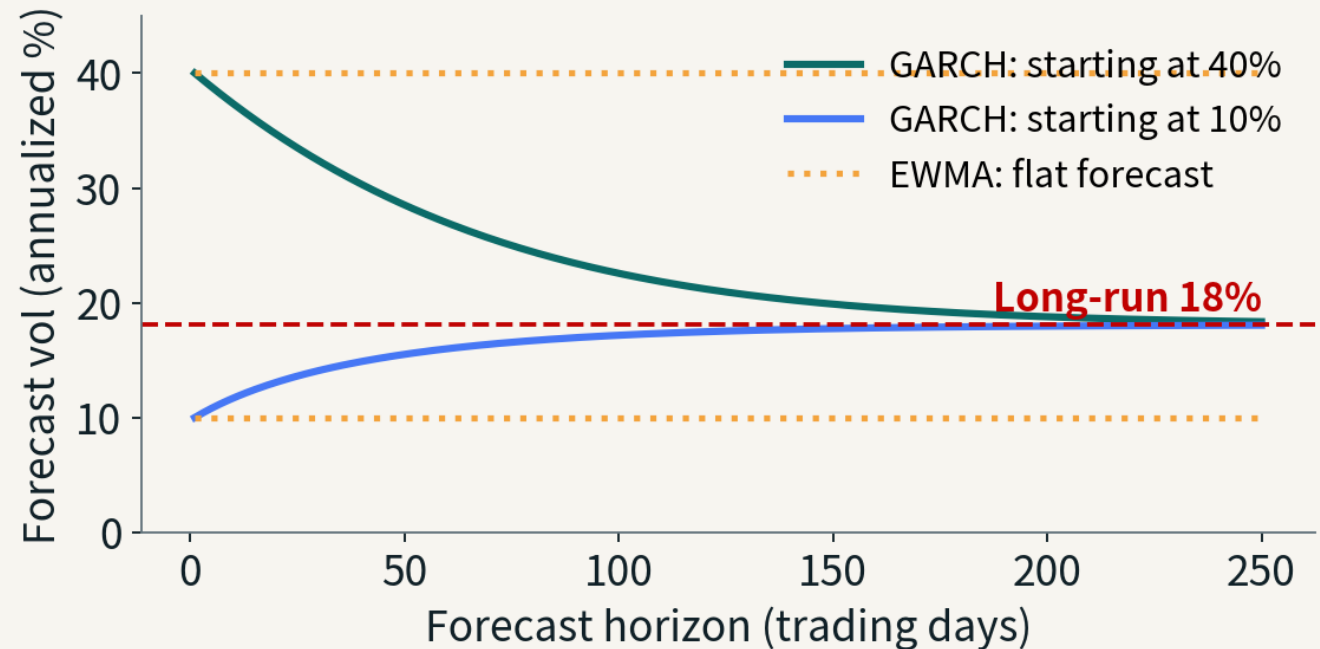
Forecast paths using S&P 500 estimates ($\alpha + \beta = 0.981$), annualized

Mean reversion

Same destination

- Half-life ≈ 35 days
- EWMA stays flat

GARCH wins at long horizons



Estimate by maximum likelihood, but test for ARCH effects first

Assuming $z_t \sim N(0,1)$, find ω, α, β that maximize the log-likelihood

① **Maximum likelihood (MLE) — solved by numerical optimization**

$$\ln L(\omega, \alpha, \beta) = -\frac{1}{2} \sum_{t=1}^T \left[\ln 2\pi + \ln \sigma_t^2 + \frac{\varepsilon_t^2}{\sigma_t^2} \right]$$

② **Engle's ARCH-LM test — do past squared residuals explain today's?**

$$\varepsilon_t^2 = c_0 + c_1 \varepsilon_{t-1}^2 + \cdots + c_q \varepsilon_{t-q}^2 + u_t, \quad LM = T \cdot R^2 \sim \chi_q^2$$

S&P 500: $LM(5) = 1,687$, $p < 0.001$ — a clear ARCH effect justifies using GARCH

KOSPI case: the gap between GARCH and implied volatility is information

Course example — GARCH(1,1) fitted to KOSPI daily data to forecast 30-day average volatility

Long-run vol

13.20%

V_L , annualized

30-day forecast

13.18%

Now $\approx$ long run

VKOSPI

10.71%

Implied vol

Gap

≈ 2.5 pp

Market calmer

GARCH uses past returns only; VKOSPI reflects what market participants expect

The gap is a starting point for judging whether options are rich or cheap

※ KOSPI spreadsheet exercise as of May 2014 (original course slide 21). Hand calculations are in the workbook

04

GARCH vs. EWMA

Same ingredients, but only one of them is pulled back to a long-run level

MA, EWMA and GARCH differ only in how they weight past squared returns

All three are weighted averages of past r^2

Moving average (MA)

$$\sigma_t^2 = \frac{1}{m} \sum_{i=1}^m r_{t-i}^2$$

Params	m only
Weights	equal, m days
Forecast	flat

EWMA

$$\sigma_t^2 = (1 - \lambda) r_{t-1}^2 + \lambda \sigma_{t-1}^2$$

Params	λ only
Weights	geometric
Forecast	flat

GARCH(1,1)

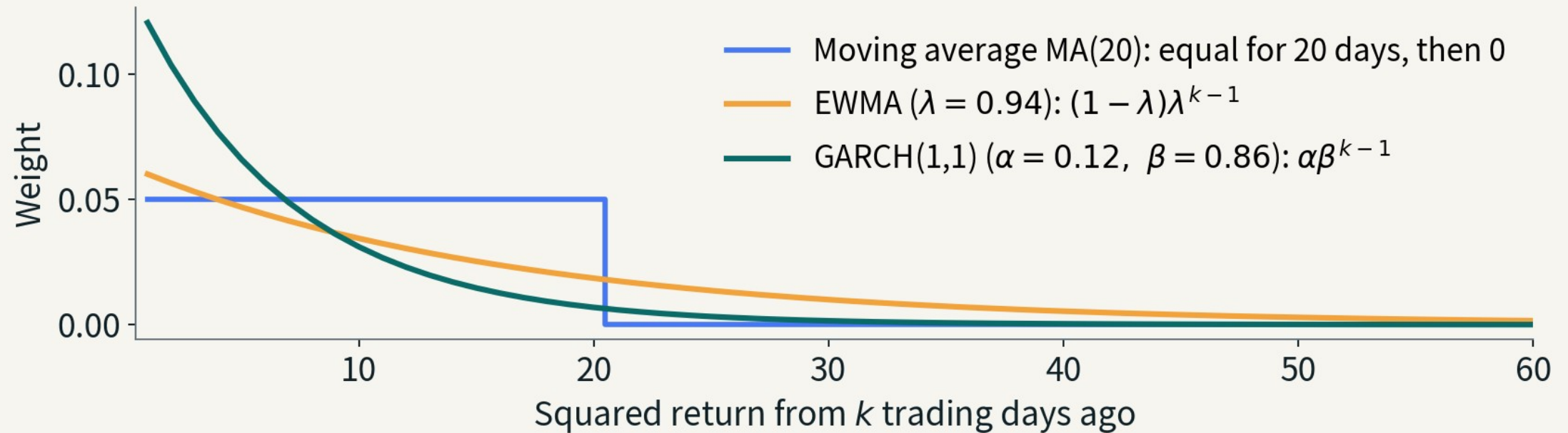
$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$$

Params	ω, α, β
Weights	geometric
Forecast	reverts

One difference matters: only GARCH has an anchor — the long-run variance (ω)

Plotting the weights shows the difference at a glance

Weight on the squared return from k days ago — GARCH uses S&P 500 estimates



※ MA weights fall off a cliff at day 20, so volatility drops abruptly when a big shock leaves the window

EWMA is GARCH with the long-run variance term removed

Rewrite GARCH with long-run variance V_L and weight γ , and the link appears

GARCH(1,1) in long-run-variance form ($\omega = \gamma V_L$)

$$\sigma_t^2 = \gamma V_L + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \gamma + \alpha + \beta = 1$$

Conditions that turn GARCH into EWMA

$$\omega = 0, \quad \alpha = 1 - \lambda, \quad \beta = \lambda \quad \implies \quad \alpha + \beta = 1, \quad \gamma = 0$$

With $\alpha + \beta = 1$, shocks never fade — this is IGARCH (integrated GARCH)

EWMA has no long-run level to return to, so its forecast is a flat line

The same +10% shock updates the two models differently

Course example — yesterday's vol 2.00%, yesterday's return +10%, long-run vol 2.00%

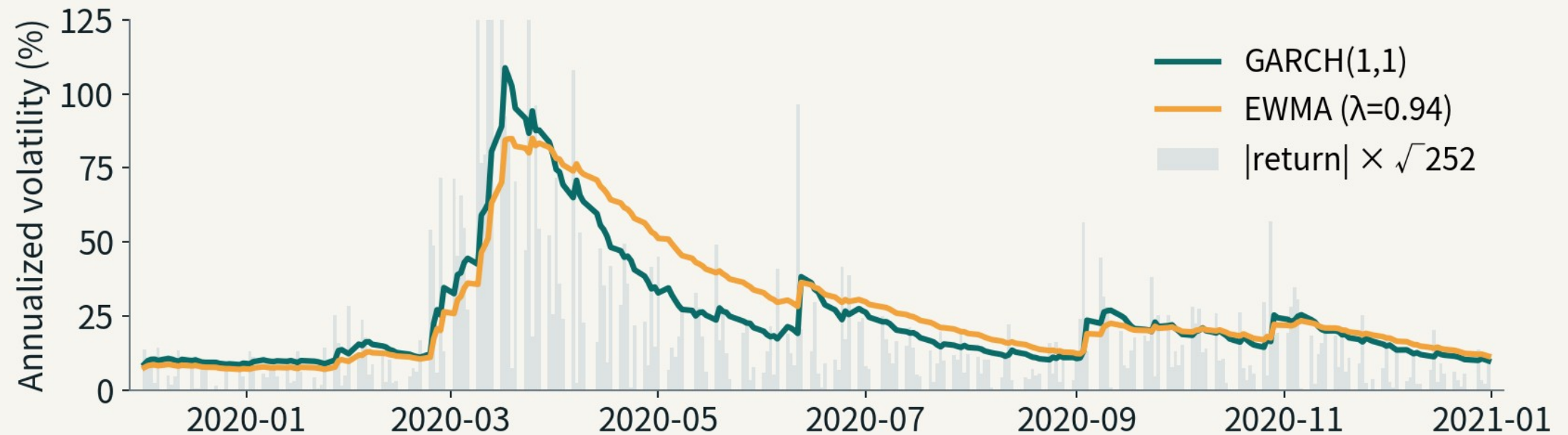
	EWMA	GARCH(1,1)
Parameters	$\lambda = 0.94$	$\alpha = 0.03, \beta = 0.94, \gamma = 0.03$
Weight on yesterday's shock	$1 - \lambda = 0.06$	$\alpha = 0.03$
Weight on long-run variance	0	$\gamma = 0.03$
Today's volatility	3.12%	2.62%

$\alpha (= 1 - \lambda)$ sets the size of the reaction; the long-run anchor sets the path afterwards

※ Step-by-step hand calculations are in the workbook

In COVID both estimates moved together but peaked differently

S&P 500, Dec 2019 – Dec 2020, annualized — grey bars show $|\text{daily return}| \times \sqrt{252}$



※ Peaks: GARCH $\approx 109\%$, EWMA $\approx 85\%$ — GARCH, with a larger α , rose faster and fell faster

Industry chose EWMA for simplicity rather than accuracy

J.P. Morgan RiskMetrics (1994) — a single $\lambda = 0.94$ for every asset

EWMA (RiskMetrics)

No estimation

- Thousands of assets at once
- Fine for short-term VaR

Weak at long horizons



GARCH(1,1)

Fitted per asset

- Reverts to the long run
- Forecasts by maturity

Suits options, long-run risk

Default: EWMA for 1- to 10-day VaR; GARCH for horizons beyond a month and for options

05

GARCH vs. Stochastic Volatility

Is volatility fixed by past data, or does it fluctuate on its own? Plus key extensions

Both models have four parameters but different sources of volatility

In SV, log volatility h_t follows an AR(1) with its own shock η_t

GARCH(1,1) — parameters $\mu, \alpha_0, \alpha_1, \beta_1$

$$y_t = \mu + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_t^2), \quad \sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

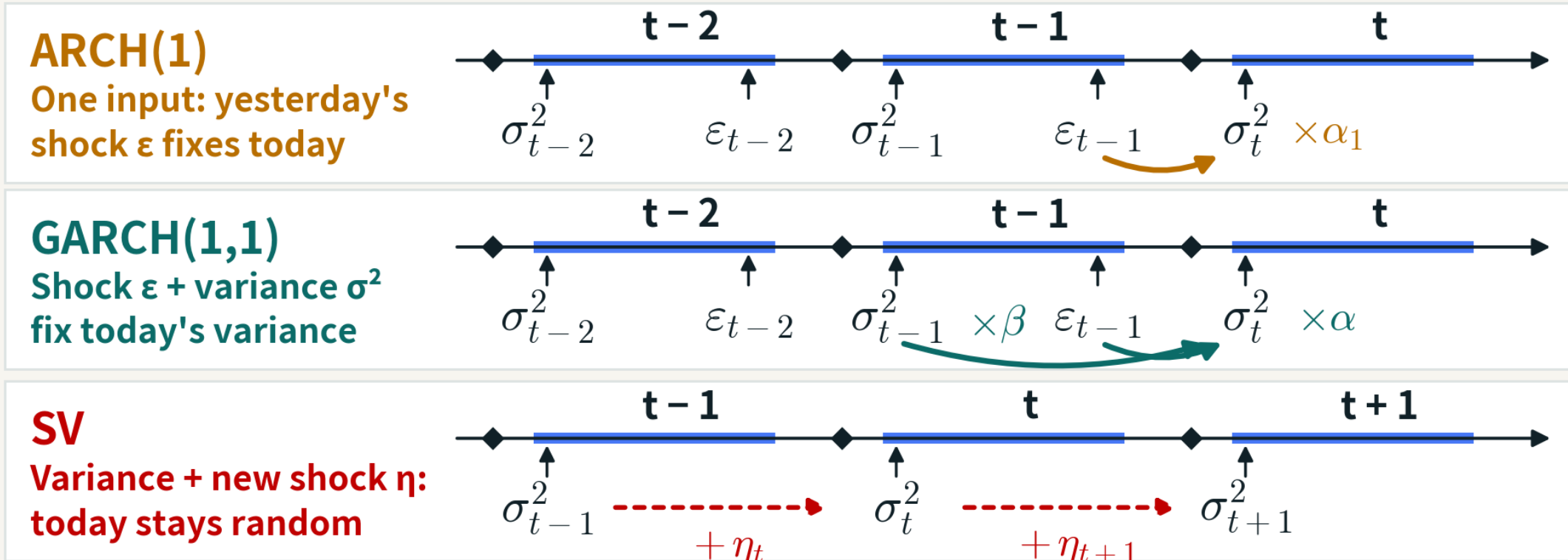
Standard SV — parameters μ_h, ϕ, ω_h^2 (+ mean μ)

$$y_t = \mu + e^{h_t/2} z_t, \quad h_t = \mu_h + \phi (h_{t-1} - \mu_h) + \eta_t, \quad \eta_t \sim N(0, \omega_h^2)$$

GARCH: variance is driven by the observed return ε_{t-1}

SV: variance is driven by a separate, unobserved shock η_t

On a timeline, ARCH/GARCH variance is set a day early; SV variance is not



※ Blue bar = trading day, ◆ = market close, short arrow = when a value becomes known, curve = inputs to today's variance

This difference decides estimation: MLE for GARCH, filtering and MCMC for SV

If variance is a function of observed data the likelihood is explicit; if latent, it must be integrated out

GARCH

$$\sigma_t^2 \in \mathcal{F}_{t-1}$$

Variance	computed
Formula	closed form
Method	one-shot MLE

SV

$$\sigma_t^2 \notin \mathcal{F}_{t-1}$$

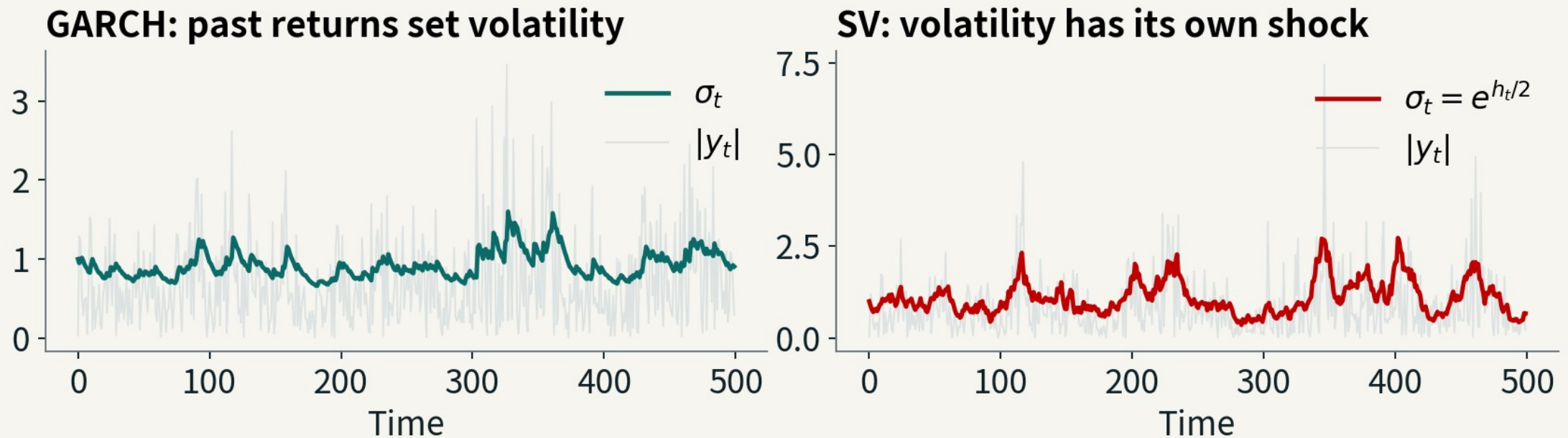
Variance	latent
Formula	high-dim integral
Method	Kalman · MCMC

SV is more flexible but computationally heavy; GARCH is less flexible but instant

※ **SV: $\log(y^2)$ transform, normal-mixture approximation, then Kalman filter or MCMC (Kim, Shephard & Chib, 1998)**

SV describes a world where volatility moves independently of returns

Simulation — GARCH ($\omega = 0.05$, $\alpha = 0.10$, $\beta = 0.85$) vs. SV ($\phi = 0.95$, s.d. of $\eta_t = 0.25$)



※ Left: σ_t jumps right after a large $|y_t|$. Right: σ_t rises and falls on its own, regardless of returns

GJR-GARCH captures the leverage effect: bad news raises volatility more

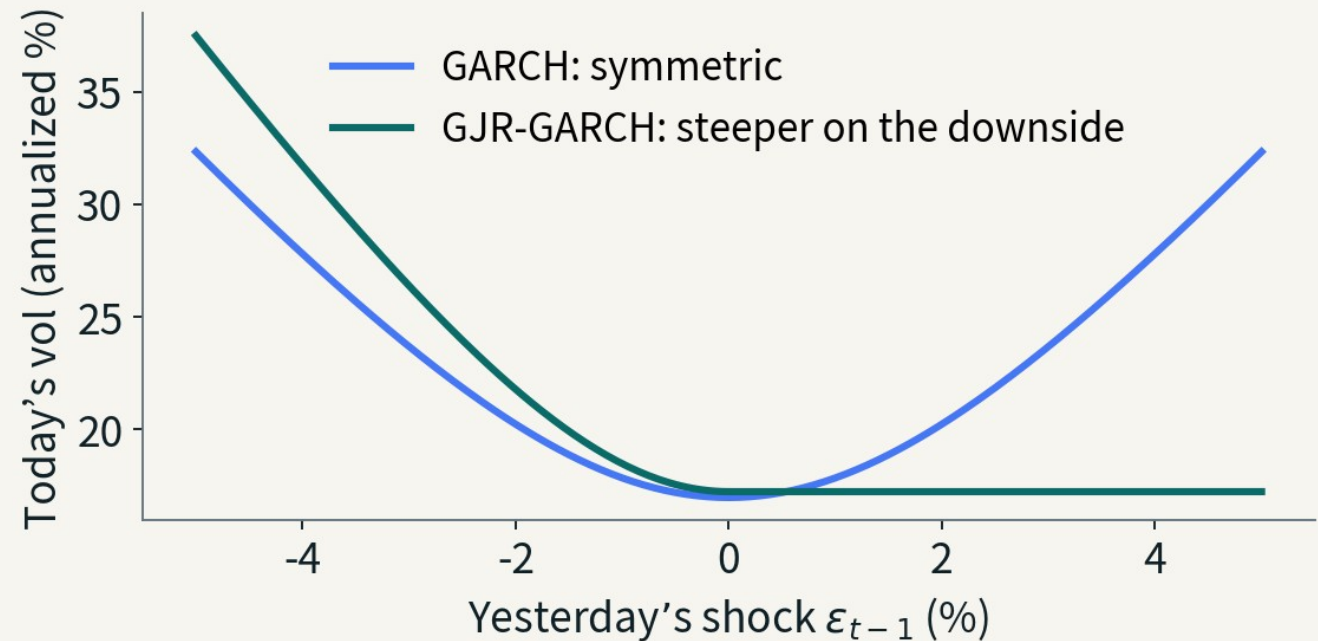
News impact curve — today's volatility by size of yesterday's shock, S&P 500 estimates

GJR estimates

$$\gamma = 0.18, \alpha \approx 0$$

- Only down days add vol
- Up-day response ≈ 0

Bad news rings louder



Small changes to the GARCH equation capture asymmetry and persistence

GJR (1993), EGARCH (1991), IGARCH — three representatives of many GARCH variants

GJR-GARCH — reacts γ more on down days ($I = 1$)

$$\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad I_{t-1} = \mathbf{1} [\varepsilon_{t-1} < 0]$$

EGARCH — models log variance, so no sign constraints are needed

$$\ln \sigma_t^2 = \omega + \alpha (|z_{t-1}| - E|z|) + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2$$

IGARCH: $\alpha + \beta \approx 1$. KOSPI (2000–2026) estimates 0.996, a half-life of about 160 days

Like a unit root, shocks linger and long-run forecasts look like EWMA

The four models differ in their inputs and where forecasts end up

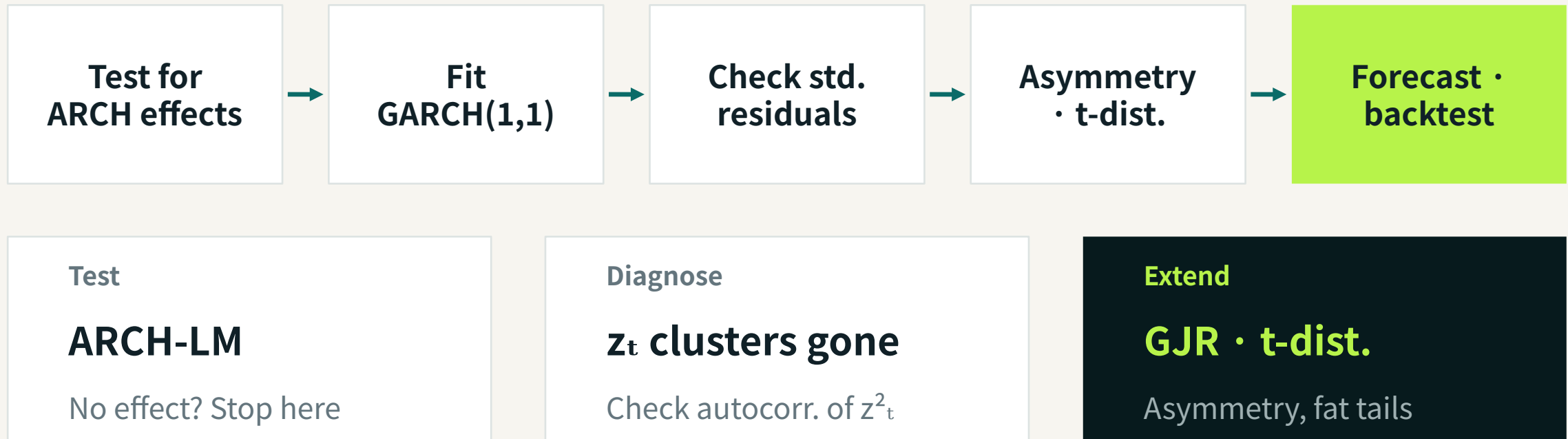
One-page summary — reading down the rows traces the models' evolution

Model	Variance inputs	Variance params	Long-run forecast	Estimation
ARCH(q)	past q days of ε^2	q + 1	long-run var.	MLE
GARCH(1,1)	yesterday $\varepsilon^2 + \sigma^2$	3	long-run var.	MLE
EWMA	yesterday $r^2 + \sigma^2$	λ only	flat	fixed λ
SV	yesterday h + shock η	3	long-run mean	MCMC · filter

The starting point is always GARCH(1,1)

In practice, start from GARCH(1,1) and extend only as needed

Standard workflow for volatility modeling



Use a complex model only once you know why the simple one failed

**We cannot call the direction of returns,
but their size clusters and remembers.**

GARCH sums up that memory in three numbers.

Next step
Try the workbook

Sources: S&P 500 and KOSPI daily closes (Yahoo Finance); original course materials (ARCH/GARCH notes, LaBarr lecture slides, Bayesian econometrics textbook Figs. 6.11 and 9.9)